Supplementary Evidence for From Neighbor Samples to Predictions

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S1 Overview and Evidence Roles

This supplement makes the complete evidence readable without duplicating every table in the main paper. It contains the detailed mathematical conditions and native-operator contracts; the original protocol, checkpoint roles and all 15 label-free calibration rows; the complete middle-budget mean table and all displayed final per-seed endpoints; paid exact/RAM reuse; historical mechanisms and correction controls; and the reproducibility/accounting boundary. The main paper presents three complementary figures for restoration, estimator error versus decisions, and quality–time–memory. Its result values are unchanged by this reorganization.

The completed final study comprises five fixed groups, each with seeds 11, 23 and 37: HGCN, HyboNet, GraphSAGE and GAT on ogbn-arxiv, plus GraphSAGE on PubMed. Both samplers, all frozen budgets, every prescribed repeat and negative or infeasible outcome remain represented. The 24-checkpoint historical synthesis is fit/development evidence under its original training regimes. It is not relabeled as independent test evidence. Exact output is a computation reference, not a label oracle. Different native error and margin scales are compared only within a model.

The tables below transcribe already accepted descriptive scalar exports. No label, checkpoint tensor or raw mechanism array is read for this document; no statistic is refit and no prediction is rescored. The source tables are already aggregated at the stated seed/repeat level. Original-precision CSVs remain adjacent in `reproducibility/final-accepted-scalars/`; the readable source is `reproducibility/final-per-seed-results.md`. Repeated node–plan events do not create independent training seeds. This is an internal evidence supplement, not a completed public artifact: authors, public checkpoint access and native-source redistribution/reconstruction remain outstanding.

How to read dated records. Historical preparation files and superseded source receipts retain earlier "pending" phrases as provenance. They do not override the completed final collection and accepted summaries used here. The current mathematics-to-implementation map and literature/formula attribution table reflect the completed evidence and its conditional scope. All 15 final cells and the complete scalar synthesis are complete. Historical tables based on a different subset or earlier scope are not substituted for the present results. The detailed source audit establishes implementation correspondence, not untested numerical domain certificates or a universal task-quality theorem.

S2 Interventions and Decision-Level Analysis

S2.1 A four-corner intervention

The message-passing abstraction follows the established MPNN framework [3]; its use here does not introduce a new general model family. Let G, features, and trained parameters be fixed. At a chosen layer, let X be the incoming messages from the exact prefix and $\widehat{X}$ those from the sampled prefix. Write $f_b(X)$ for this layer's native computation, where $b = 0$ uses the specified sampled neighborhood and $b = 1$ uses all neighbors. Thus f_b includes message weighting, normalization, and the relevant activation, rather than just a matrix product. The four states are

$$H_{00} = f_0(\widehat{X}), \qquad H_{10} = f_1(\widehat{X}), \tag{S1}$$

$$H_{01} = f_0(X), \qquad H_{11} = f_1(X). \tag{S2}$$

The first subscript indicates restoration of local computation; the second indicates restoration of upstream representations. Sampling randomness is paired whenever the operation is sampled. At the final message-passing layer, the original task head maps each state to scores. At an earlier layer, the original suffix is recomputed using the registered paired sampling plan. Intervening at one node without updating its affected descendants would define a different experiment.

For a fixed attention operator, "restore local" means recomputing full-neighborhood attention on the *same incoming messages*; it does not replace sampled attention weights with weights computed from another upstream state. Attention's dependence on X makes the functional definition in Equation (S2) preferable to assuming one constant aggregation matrix. The primary new intervention is fixed at the final message-passing layer before new effects are inspected.

Using the standard two-factor interaction framework [5], with 0/1 baseline differences rather than a ± 1-coded effect normalization, for scalar quality Q define

$$\Delta_{\text{local}} = Q_{10} - Q_{00}, \tag{S3}$$

$$\Delta_{\text{up}} = Q_{01} - Q_{00}, \tag{S4}$$

$$\Delta_{\text{joint}} = Q_{11} - Q_{00}, \tag{S5}$$

$$I_Q = Q_{11} - Q_{10} - Q_{01} + Q_{00}. \tag{S6}$$

Then $\Delta_{\text{joint}} = \Delta_{\text{local}} + \Delta_{\text{up}} + I_Q$ exactly. A nonzero I_Q records nonadditivity on this metric and intervention scale. It is not evidence for a particular causal mechanism by itself, and the three improvements are not independent percentages of a total error source. Accuracy is discontinuous and can be nonadditive even when an intermediate representation decomposes linearly.

S2.2 Finite source decomposition and propagation

For linear aggregation of fixed messages, exact matrices A and sampled matrices $\widehat{A}$ give the bookkeeping identity

$$\begin{aligned}\widehat{A}\widehat{X} - AX = (\widehat{A} - A)X + A(\widehat{X} - X) \\ + (\widehat{A} - A)(\widehat{X} - X).\end{aligned} \tag{S7}$$

Equation (S7) is direct algebra, not a new decomposition theorem. The terms correspond to a local operator change, an inherited message change, and a cross term relative to this reference. Squared norms retain cross inner products. Even a zero conditional mean for one term does not justify discarding its correlations with other layers. For attention, one may instantiate matrices at their

actual endpoint inputs and still obtain this algebraic identity; its terms then mix weight adaptation and sampling. They are not equivalent to the functional interventions in Equation (S2) unless the relevant dependencies are held fixed.

A finite formulation avoids assuming that a first-order approximation dominates. Let the exact layer map be F_ℓ and the sampled one $\widehat{F}_\ell$, with $\delta_\ell = \widehat{H}_\ell - H_\ell$. Define the local residual on actual incoming states as

$$e_\ell = \widehat{F}_\ell(\widehat{H}_{\ell-1}) - F_\ell(\widehat{H}_{\ell-1}). \tag{S8}$$

Then, for any selected linear map J_ℓ,

$$\begin{aligned} \delta_\ell &= J_\ell \delta_{\ell-1} + e_\ell + r_\ell, \\ r_\ell &= F_\ell(\widehat{H}_{\ell-1}) - F_\ell(H_{\ell-1}) - J_\ell \delta_{\ell-1}. \end{aligned} \tag{S9}$$

With identical input features, recursive substitution gives

$$\delta_L = \sum_{\ell=1}^{L} \left(\prod_{j=L}^{\ell+1} J_j \right) (e_\ell + r_\ell). \tag{S10}$$

The ordered empty product is the identity. These are exact identities with the stated residuals, not small-error guarantees. On a differentiable, common coordinate branch, choose $J_\ell = DF_\ell(H_{\ell-1})$. If the entire segment between $H_{\ell-1}$ and $\widehat{H}_{\ell-1}$ lies in that branch and the Jacobian obeys $\|DF_\ell(u) - DF_\ell(v)\| \le K_\ell \|u - v\|$ along it, integrating the Jacobian difference gives the standard Taylor bound [6]

$$\|r_\ell\| \le \frac{K_\ell}{2} \|\delta_{\ell-1}\|^2. \tag{S11}$$

This conditional bound is classical; the experiment does not certify its smoothness assumptions for every encountered state. ReLU boundary changes, clipping, nonadmissible manifold coordinates, and numerical error need their own treatment. In manifold-valued layers, subtracting ambient coordinates is permitted as bookkeeping but is not automatically an intrinsic distance.

The measurable implication is narrow: a final layer can have $e_L = 0$ while $\delta_L \neq 0$ because its incoming representations differ. A one-message-layer model with identical raw inputs has no sampled upstream message-passing prefix. It is a structural boundary case, not a failed replication of an upstream effect in a deeper model.

S2.3 Why smaller local error need not improve accuracy

Let s_c be a class score, y the true label, and

$$m_y = s_y - \max_{c \neq y} s_c = \min_{c \neq y} M_{yc}, \qquad M_{ab} = s_a - s_b. \tag{S12}$$

Correctness is determined by signed margins and the declared tie rule. For a linear head $s_c(h) = w_c^\top h + b_c$, direct subtraction for a finite representation change d gives the elementary identity

$$M_{ab}(h + d) - M_{ab}(h) = (w_a - w_b)^\top d. \tag{S13}$$

Error magnitude alone therefore does not determine the sign of the margin change. For example, with a reference binary margin 0.1 and head difference $(1, 0)$, the error $d_1 = (-0.2, 0)$ changes the prediction, while the larger error $d_2 = (0, 1)$ leaves its margin unchanged. This elementary construction concerns decisions of a fixed head, not the expected behavior of a trained architecture.

If $m_y(h) > 0$, a sufficient condition for preserving the correct prediction is

$$\max_{c \neq y} \left| (w_y - w_c)^\top d \right| < m_y(h). \tag{S14}$$

The norm bound $\|w_y - w_c\|_* \|d\|$ is sufficient but may be loose because it discards direction. Reducing this bound need not change any label. Conversely, moving across a narrow decision boundary can change a label at small displacement. A reference full prediction can itself be wrong; agreeing with it and improving label accuracy are different objectives.

For each intervention we record corrected events C (sampled wrong to intervention correct) and harmed events H (sampled correct to intervention wrong). On the same N node–plan events,

$$\mathrm{Acc}_{\mathrm{intervention}} - \mathrm{Acc}_{00} = \frac{C - H}{N}. \tag{S15}$$

Corrections and harms reveal offsetting changes that a net gain can conceal. The archived development synthesis additionally reports changed predictions for which both endpoints are wrong, using its retained class predictions; correctness flags alone do not identify those events in the new summary. True-label margins, full-winner margins, and representation errors remain separately named quantities.

S2.4 Sampling guarantees depend on the estimator

For fixed messages x_j and a fixed linear aggregation target $a = \sum_j a_j x_j$, the Horvitz–Thompson estimator [4] with inclusion indicators I_j and positive inclusion probabilities π_j is

$$\widehat{a} = \sum_j I_j \frac{a_j x_j}{\pi_j}, \qquad \mathbb{E}[\widehat{a} \mid X] = a. \tag{S16}$$

Its covariance depends on joint inclusion probabilities π_{jk}, not only marginal probabilities. Uniform sampling without replacement with a fixed, nonzero sample size gives an unbiased sample mean for fixed messages. The LABOR-0 bridge, distinguished from the estimator guarantee in the original LABOR work [1], accepts no extra importance weights: a nonempty returned importance-weight array is rejected. Its GraphSAGE mean divides the sampled message sum by the realized random neighbor count and returns a zero neighbor aggregate when that count is zero. Random count and empty-neighborhood events prevent inheriting unconditional unbiasedness from the fixed-count Uniform statement. Historical HyboNet SUM uses weights $A_{ij}/\min(1, k/d_i)$ on its specified eligible neighbors, which is a different estimator contract. For correlated layer sampling, neither conditional independence across messages nor across layers may be assumed. Every estimator must be mapped to its actual self-message, duplicate, and empty-neighborhood handling before Equation (S16) is invoked. If a shared random seed correlates the upstream state and current plan, conditioning on the state can change the sampling law.

Attention is a crucial exception to a naive extension. For fixed inputs, define full attention values z_j and positive unnormalized weights $q_j = \exp(e_j)$. The output is $a = N/D$, where $N = \sum_j q_j z_j$ and $D = \sum_j q_j$. Even if probability weighting makes $\widehat{N}$ and $\widehat{D}$ separately unbiased, their ratio generally is not. Writing $\widehat{N} = N + \delta N$, $\widehat{D} = D + \delta D > 0$ gives the exact identity

$$\frac{\widehat{N}}{\widehat{D}} - \frac{N}{D} = \frac{\delta N - a\,\delta D}{D + \delta D}. \tag{S17}$$

For centered perturbations $\mathbb{E}[\delta N] = \mathbb{E}[\delta D] = 0$, applying the standard delta method [7], with the denominator bounded away from zero, suitable concentration and controlled expected remainder,

gives the approximate second-order mean term $(a\,\mathrm{Var}(\delta D) - \mathrm{Cov}(\delta N, \delta D))/D^2$. Without centering, first-order mean terms also remain. The approximation is not a finite-sample unbiasedness claim. Native sampled-neighborhood softmax and a probability-weighted ratio can also be different operators; the chosen implementation must be declared before effects. Applying a local variance result to post-softmax output or accuracy without this mapping would change the mathematical object being analyzed.

S2.5 A Lorentz-head specialization

The bound implementation of the fully hyperbolic architecture [2] first forms the weighted ambient sum m of transformed messages, then returns $h = m/\rho(m)$, where

$$\rho(m) = \sqrt{\max\{|\langle m, m\rangle_L|, 10^{-8}\}}. \tag{S18}$$

Its native prototype score is $s_c(h) = 2 + 2\langle h, p_c\rangle_L + b_c$, with no subsequent projection, distance clipping, or softmax in the decoder. The decoder uses a fixed unit-radius Lorentz manifold; its class prototypes are manifold parameters with projected initialization, but decoding does not recheck or project loaded prototypes. We therefore separate the following real-arithmetic identity from an empirical certificate of unit-norm checkpoint values. For fixed finite $D_{ab} = p_a - p_b$ and $\beta_{ab} = b_a - b_b$, the positive scale ρ preserves the sign of the pairwise margin:

$$\psi^{\rho}_{ab}(m) = \rho(m) M_{ab}(h) = 2\langle m, D_{ab}\rangle_L + \rho(m)\beta_{ab}. \tag{S19}$$

For finite m, d in this domain,

$$\begin{aligned}\psi^{\rho}_{ab}(m+d) - \psi^{\rho}_{ab}(m) = {} & 2\langle d, D_{ab}\rangle_L \\ & + [\rho(m+d) - \rho(m)]\beta_{ab}.\end{aligned} \tag{S20}$$

This follows by substitution, without linearization or a unit-prototype assumption. If every class bias is equal, the positive final normalization scale alone does not alter the ordering of a *fixed* aggregate. Unequal biases make the scale explicitly task relevant. This does not make preceding aggregation or earlier normalizations harmless. Floating-point evaluation can leave a finite arithmetic residual; the identity is not a bitwise-equality claim.

If m is future-pointing timelike and $-\langle m, m\rangle_L \geq 10^{-8}$, then $\rho(m) = \nu(m) = \sqrt{-\langle m, m\rangle_L}$. Substituting ν at both endpoints of Equation (S20) requires these conditions at both m and $m + d$. With unit future-pointing prototypes as well, $2 + 2\langle h, p_c\rangle_L$ is the negative chordal distance proxy $-(-2 - 2\langle h, p_c\rangle_L)$, rather than negative squared geodesic distance. Outside that domain, Equation (S20) still describes this native scalar-normalization head in real arithmetic, but does not establish hyperbolic distance semantics. Historical native FP32 records include nonunit intermediate messages and violations of idealized intermediate conditions. Thus domain coverage and arithmetic residuals require their own evidence. The factorization does not transfer unchanged to HGCN's linear decoder, a Poincaré distance head, or another curvature convention; model-level comparisons also cannot isolate curvature from depth, head, or normalization.

S3 Workload-Matched Cost and Selection

Let T be an ordered target-node workload with a fixed batching rule. A deployable sampling request includes CPU neighborhood planning, construction of the required blocks, host–device movement, and synchronized model computation. We distinguish

$$t_{\mathrm{request}} = t_{\mathrm{plan}} + t_{\mathrm{prepare}} + t_{\mathrm{transfer}} + t_{\mathrm{forward}} + t_{\mathrm{other}}, \tag{S21}$$

only when the components are nonoverlapping intervals of the same clock. If instrumentation overlaps or clocks differ, components are reported separately rather than summed. Graph preprocessing, loading, format construction, and model loading are one-time costs with their own scope. Request latency distributions and aggregate workload wall time answer different questions and are both retained.

Same fanout is not the same cost. Layer-level sharing, node reuse, duplicates, attention calculation, and block preparation can differ between samplers at the same fanout. Three budgets are therefore fixed using label-free timing and a predetermined discrete candidate rule. All realized cost deviations are reported. Points on an observed quality–time–memory curve are not treated as a continuous interpolated algorithm: an equal-time comparison is valid only at a measured feasible operating point or a prospectively specified randomized mixture, which this round does not add.

Exact outputs and reuse. For q repetitions of the same target workload, let S_s be the sampled route's setup, $T_s(T)$ its per-workload cost, S_f the exact route's setup, P_f full-output precomputation, and $T_c(T)$ the cached query cost. Then

$$C_s(q) = S_s + qT_s(T), \tag{S22}$$

$$C_c(q) = S_f + P_f + qT_c(T). \tag{S23}$$

By elementary comparison of these accounting expressions, when $T_s(T) > T_c(T)$ caching is cheaper for

$$q > \frac{S_f + P_f - S_s}{T_s(T) - T_c(T)}. \tag{S24}$$

The smallest valid integer is obtained from the strict inequality, and negative setup differences are retained. If the denominator is nonpositive, this inequality does not give the stated crossover. Host output bytes, GPU working memory, and invalidation after graph or parameter changes are part of the comparison. A whole-graph full forward cannot be compared directly with one small sampled target batch; both routes must serve the same total requested outputs.

Decision rule and reporting. For each frozen model, seed, and workload, report the measured discrete frontier over the two samplers and three budgets, together with the exact and amortized-cache routes. A method is dominated only when another measured feasible point is no worse in the reported quality, time, and memory dimensions under the same accounting scope. Near-equal-quality tolerance and feasibility thresholds are frozen before the new final evaluation. Development outcomes may inform the written interpretation, but cannot change the set of final methods, checkpoints, or budgets.

S4 Experimental Protocol and Evidence Boundaries

S4.1 Assets and bounded comparison

The completed study reuses the 24-checkpoint historical captures and 12 arxiv base checkpoints, and adds three PubMed base trainings. Arxiv uses the original plain HGCN, HyboNet, GraphSAGE and GAT models, before the later 50-epoch correction adaptations. Each new group retains seeds 11, 23 and 37; all 15 final cells are complete. Weights were fixed by role and provenance, not correction gains. The supplementary inventory records every checkpoint's hash, training source, selection rule, split and regime, separating historical fit evidence from final evaluation.

The 24-checkpoint collection crosses two hyperbolic implementations, two datasets, two training regimes, and three seeds. Airport uses budgets 4/8/16; Disease-NC uses 1/2/4, with full-neighborhood references. The historical HGCN implementation has one message-passing layer. HyboNet has five on Airport and three on Disease-NC. Disease-NC has only 42 validation nodes, compared with 476 on Airport. In contrast, the new arxiv HGCN and HyboNet checkpoints each have three message layers, as do the new ordinary-model checkpoints. In particular, the historical HGCN upstream branch is a structural alias; this is not an assumption about the new three-layer HGCN. Equal depth alone does not match the models' heads, normalization or training. These facts constrain interpretation rather than being changed to make model effects look uniform.

For the new arxiv study, the only samplers are Uniform and LABOR. All use the same target IDs, target ordering, batch specification, and task metrics. Three operating budgets were calibrated without labels; the major mechanism intervention was fixed at the final message-passing layer at the middle budget. Each sampler's actual probabilities, weighting, self loops, empty-neighborhood rule, and random namespaces were frozen in the source contract. Existing HGCN and HyboNet weights were reused for the required comparable frozen tests. The second dataset was chosen using checkpoint availability and documented graph differences, not which dataset previously produced favorable effects.

The final target workload contains all 48,603 arxiv test nodes and all 1,000 PubMed test nodes. With fixed batch size 128, one workload is 380 arxiv requests (379 full batches and a final batch of 91) or eight PubMed requests (seven full batches and a final batch of 104). Each of the four repeats uses the same complete target workload. Calibration instead uses the fixed development targets specified below; its request medians are not final-workload times.

Fixed budgets and repeats. Uniform uses the same fanout at each layer: 5/15/30 on arxiv and 1/4/8 on PubMed. For each model, one seed-11 label-free calibration uses eight fixed batches of at most 128 targets, three timed repeats and one paid warm-up. LABOR-0 uses no external sampling-probability input and zero importance-sampling iterations. Its arxiv candidate fanouts are 5, 8, 12, 15, 20, 24, 30, 36, 48, 64, and 80; PubMed uses 1, 2, 4, 6, 8, 12, and 16. For each Uniform tier, the rule selects the largest LABOR fanout whose median over 24 request timings does not exceed the Uniform time; other seeds inherit that lock. If none qualifies, the tier has no feasible cost-matched LABOR point, and the minimum fanout's actual cost is still reported. No interpolation or new sampler repairs an infeasible tier. New evaluation uses four paired repeats per seed and tier. Near-equal accuracy means no seed is more than 0.2 percentage points below the comparison point; all actual differences remain visible.

All five label-free A800 cost locks are accepted (Table S1); no accuracy labels were evaluated for this selection. SAGE/arxiv's low-tier minimum LABOR point exceeds its fixed target by about 0.298%, so that tier is explicitly infeasible under the no-over-target rule. It is neither recalibrated nor treated as an achieved equal-cost match. On PubMed, all three Uniform tiers select the same LABOR $K = 6$ point: these are three comparisons against fixed Uniform endpoints, not three distinct LABOR operating points. Uniform $K = 8$ is measured faster than $K = 4$ on that workload. We retain this nonmonotonicity rather than smoothing the costs or repeating calibration to obtain monotone points. The final workload's actual costs and deviations must still be reported. These discrete label-free measurements establish frozen budgets, not a quality–cost benefit. Cross-dataset absolute times do not compare identical target workloads.

Diagnostic versus deployment computation. The middle-budget four-corner intervention uses frozen whole-graph layerwise sampling plans to construct each sampled prefix, followed by the

Table S1: Frozen label-free A800 calibration, seed 11; seeds 23/37 inherit each lock. Times are median request milliseconds over 24 timings per point. These are budget-selection observations, not official-test quality or final speedup results. The infeasible minimum-cost point is retained.

Dataset / model	Tier	Uniform K	Uniform ms	LABOR K	LABOR ms	Feasible?
ogbn-arxiv / HGCN	low	5	70.151	8	61.559	Yes
ogbn-arxiv / HGCN	middle	15	113.843	12	95.815	Yes
ogbn-arxiv / HGCN	high	30	250.854	36	226.308	Yes
ogbn-arxiv / HyboNet	low	5	41.329	5	37.245	Yes
ogbn-arxiv / HyboNet	middle	15	98.561	15	93.485	Yes
ogbn-arxiv / HyboNet	high	30	223.856	36	217.241	Yes
ogbn-arxiv / GraphSAGE	low	5	35.030	5	35.134	No
ogbn-arxiv / GraphSAGE	middle	15	95.080	15	83.924	Yes
ogbn-arxiv / GraphSAGE	high	30	215.652	36	205.791	Yes
ogbn-arxiv / GAT	low	5	43.810	5	41.547	Yes
ogbn-arxiv / GAT	middle	15	100.699	15	92.426	Yes
ogbn-arxiv / GAT	high	30	227.070	30	189.839	Yes
PubMed / GraphSAGE	low	1	33.723	6	33.339	Yes
PubMed / GraphSAGE	middle	4	37.158	6	33.339	Yes
PubMed / GraphSAGE	high	8	34.552	6	33.339	Yes

last-layer intervention on the complete incoming node-state arrays. This preserves the semantics of restoring local computation without filling unvisited upstream states with zeros or exact representations. Deployment quality and cost instead use target dependency-cone requests. Their baselines are named separately and are not assumed to agree bitwise or to define the same sampling distribution. Whole-graph diagnostic state caches and exact references are paid diagnostic resources, not free deployment inputs. GAT retains native sampled-neighborhood softmax and exactly one self message; a new inverse-probability attention correction is not introduced, and its output is not claimed unbiased.

Second-graph configuration. PubMed was fixed for its available qualified data, 19,717 nodes, sparse neighborhoods, and small labeled training set, which contrast with arxiv. GraphSAGE uses three layers, hidden width 256, batch normalization/ReLU, dropout 0.5, Adam with learning rate 0.01 and zero weight decay, training fanout $[4, 4, 4]$, and all 500 epochs. Its 60 official training labels are used for training. The inherited native loader uses 1,000 validation nodes (rows 60–1059), rather than the common PyG 500-node validation configuration. These 1,000 nodes are split into 500 fit and 500 development nodes by a frozen SHA256 namespace and node-ID tiebreaker; every fifth epoch, four fixed fit plans select the earliest maximum-average-accuracy checkpoint. No hyperparameter search or correction training is performed. The 1,000 official test labels were used only after the present design was frozen for its final score. Earlier hyperbolic work used this dataset's test, so we do not claim that PubMed test had never been observed; the present model/configuration was fixed without choosing it by those scores.

The three PubMed base trainings have now completed all 500 epochs and been accepted. Seeds 11, 23, and 37 select epochs 430, 220, and 500, with fit accuracies 74.30%, 75.05%, and 73.80%, respectively. These values document the original selection rule; they are not independent task-

quality results. Their allocated GPU times are 100, 87, and 82 seconds, totaling 269 seconds on NVIDIA A800-SXM4-80GB. Together with the 12 reused arxiv base checkpoints, all 15 required weights are available; none is replaced using the new results.

S4.2 Metrics, pairing, and statistical unit

Primary quality is label accuracy. For each model–seed–budget–sampler cell, we retain per-plan accuracy, true-label margin, corrections and harms relative to sampled inference, prediction mismatch, and local approximation error under a precisely named coordinate/estimator metric. Sampled and intervention outputs are paired by checkpoint, target ID, and plan. Each seed is first summarized across its prescribed plans; reported mean and sample standard deviation are then across the three training seeds. Repeated node–plan events do not create additional independent training seeds. With three seeds, intervals and sign consistency are descriptive evidence with substantial uncertainty, not high-resolution population estimates.

The four-corner table includes every seed and training regime, including zero, beneficial, non-monotone, and degenerate cases. An aggregate pooled across architectures or training regimes is labeled as such and does not replace cell-level results. In particular, $\mathrm{Acc}_{11} - \mathrm{Acc}_{00}$ may be negative because the full computation can be less accurate. The interaction in Equation (S6) is computed on the same paired set, without turning it into a fraction of independent causes.

S4.3 Development and independent evaluation

Historical fit, gate, and confirmation partitions retain their original roles. Fresh sampling plans on previously used labels do not create an independent test. Task A is an offline analysis of existing evidence and is labeled accordingly. New implementation checks and label-free calibration preceded the new scoring. Checkpoints, sampling methods, budgets, intervention locations, metrics, statistical rules, and source hashes were fully fixed before the official test was scored once. No checkpoint replacement, method tuning, budget selection, or selective rerun follows that score. A failure is retained with its execution and data scope; it is not converted into an opportunity to obtain a more favorable result.

S4.4 Resource measurements and reproducibility

Calibration and final comparisons use NVIDIA A800-SXM4-80GB; historical L40 measurements are a separate hardware stratum. Up to four jobs run on distinct ready GPUs, with serial timing within each device and recorded CPU affinity, memory, transfer and host context. Assigned-device process checks are discrete observations, not proof of continuous host isolation. Timing reports retain input residency, warm-up, synchronization, complete target workload and setup/cache scope. Allocated peaks and reserved allocator-history observations remain distinct; a mixed-budget reserved peak is not a cold minimum capacity.

The new round uses 17,299 allocated GPU-seconds: 269 for three PubMed base trainings, 311 for successful label-free calibration, 13 for the preserved GAT metadata failure, and 16,706 for the 15 final cells. All allocations are closed and released before archive. A scoped tuple/list metadata repair changed no model, sampler, checkpoint or numerical computation. Already accepted qualifications and the SAGE cost lock were inherited without repetition. Complete per-job records, failed attempts and source versions remain in the material package. Allocation, worker, scheduler and controller timings have overlapping scopes and are not added; unknown child CPU is not imputed. The final offline synthesis uses 11.375 native CPU-seconds and 14.666 wall-seconds, with no added GPU computation.

Table S2: Mathematics-to-implementation contracts, supported by scoped finite correspondence checks. Absolute local-error and native-margin scales are not comparable across models. Exact source anchors are in the supplementary map.

Object	Actual computation	Mathematical boundary
Uniform SAGE mean	Fixed-size non-self sample without replacement, realized mean, separate root transform	Conditional unbiased mean for fixed nonempty messages; no unbiased network-output or accuracy claim
LABOR-0 SAGE mean	Native empty importance-weight return, actual random sampled count, zero empty-row term	Does not inherit the fixed-count Uniform argument; positive-degree empty draws and joint inclusion matter
Hyperbolic support	Fixed self; HT-weighted non-self sum before native log/exp or Lorentz post-map	Conditional linear-estimator reasoning ends before nonlinear normalization; origin-tangent and ambient errors have different meanings
GAT attention	One self message; native per-target sampled softmax, head merge, residual and bias	Random ratio with state-dependent weights; no new inverse-probability correction or universal output-unbiasedness
Four-corner intervention	Whole-graph sampled/full prefixes; shared final sampled plan; complete-row chunks	Diagnostic and deployment baselines differ; retain interaction and pay for oracle states and support tracing
Native decisions	Native head dispatch; signed true-class margin; paired corrections and harms	Linear identities require a named affine head; Lorentz ψ requires its stated head/domain and does not describe HGCN's linear decoder

Table S2 and the detailed source map connect mathematics to the actual operators. The package retains exact hashes, environment/data provenance, all calibration locks, every accepted scalar row, per-seed tables, historical evidence and plot sources. Finite synthetic qualifications establish their stated implementation correspondence, not a task-quality guarantee. Bound source and result identities are recorded, but public checkpoint access and native-source licensing/reconstruction remain unresolved; no completed public release is claimed.

S5 Complete Final Results

The following mean table is the original middle-budget diagnostic display removed from the main paper to avoid duplicating its first figure. The per-seed tables that follow retain deployment endpoints, four-corner accuracies and native margins, corrected/introduced events, paired restoration changes, interactions, local-estimator error, and exact/RAM observations. Complete additional scalar fields and original precision remain in the exported CSVs; no favorable seed is selected.

Table S3: Final middle-budget diagnostic accuracy and restoration changes. Values are three-seed means; sampled accuracy includes sample standard deviation, with all other deviations in the supplement. Accuracy is in percent; changes/interaction are percentage points. Each seed averages four paired whole-graph-prefix plans. U: Uniform; L: LABOR. Both-restored is retained separately for each stream, including small numerical differences. These are diagnostic effects, not deployment gains or independent contribution shares.

Group	Sampler	Sampled (%)	Local Δ	Upstream Δ	Joint Δ	Interaction
HGCN / arxiv	U	70.186 ± 0.214	+0.182	+0.144	+0.283	-0.043
HGCN / arxiv	L	69.777 ± 0.312	+0.274	+0.474	+0.690	-0.058
HyboNet / arxiv	U	70.006 ± 0.196	+0.120	+0.068	+0.159	-0.029
HyboNet / arxiv	L	69.896 ± 0.173	+0.111	+0.170	+0.269	-0.012
GraphSAGE / arxiv	U	70.853 ± 0.392	+0.074	+0.053	+0.117	-0.010
GraphSAGE / arxiv	L	70.804 ± 0.453	+0.100	+0.091	+0.166	-0.025
GAT / arxiv	U	72.141 ± 0.081	+0.079	+0.091	+0.154	-0.017
GAT / arxiv	L	72.109 ± 0.102	+0.078	+0.121	+0.186	-0.013
GraphSAGE / PubMed	U	74.775 ± 0.463	-0.042	+0.275	+0.492	+0.258
GraphSAGE / PubMed	L	74.817 ± 0.753	+0.092	+0.200	+0.450	+0.158

These tables pivot accepted scalar rows only. No new forward pass, scoring, statistical aggregation, significance test or raw-array read was performed. All means, paired changes and interaction values come from the accepted exports. Decimal presentation converts fractions to percent/percentage points and bytes to MiB. Exact original precision and every additional metric remain in `final-accepted-scalars/`.

Each seed summarizes four paired repeats where applicable; three training seeds remain the replication unit. Deployment dependency-cone sampling and whole-graph-prefix diagnosis are separate streams. A positive average margin does not imply every node is correct; exact output is not a label oracle. Correction/introduced counts below are mean nodes per repeat, not unique independent cases.

Acceptance: `09bd47fbc1ad872f61c155b901a39da9d3c4600e69339d5b2b693ce8d9a36f05`. Source: metadata-v4 `ece3665a…`; frozen choices unchanged.

Frozen target denominators: 48,603 arxiv test nodes and 1,000 PubMed test nodes per repeat. With batch size128, complete deployment workloads comprise380 arxiv requests (379×128 +91) and8 PubMed requests (7×128 +104). Four repeats retain the same target set. These counts come from the accepted pre-result plan, not reconstructed labels or prediction arrays. Diagnostic transition counts below are mean nodes per repeat; do not multiply them into independent observations.

S5.1 HGCN / ogbn-arxiv

S5.1.1 Deployment: all tiers and seeds

Table S4: HGCN / ogbn-arxiv: Deployment: all tiers and seeds. Original displayed values, with all seeds retained.

Seed	Method	Tier	K	Accuracy %	Workload s	Max observed peak MiB	Feasible calibration match
11	uniform	low	5	66.9244	22.869701	95.863	reference
11	uniform	middle	15	70.1299	48.516263	302.445	reference
11	uniform	high	30	70.4144	98.042242	447.643	reference
11	labor	low	8	68.6156	27.033624	127.825	True
11	labor	middle	12	69.7385	34.621050	197.525	True
11	labor	high	36	70.3588	91.752568	433.263	True
23	uniform	low	5	66.6759	22.646078	95.881	reference
23	uniform	middle	15	69.9221	48.385256	305.111	reference
23	uniform	high	30	70.1006	98.440332	447.577	reference
23	labor	low	8	68.4191	27.100554	128.199	True
23	labor	middle	12	69.5255	34.649926	198.158	True
23	labor	high	36	70.1536	91.369919	433.465	True
37	uniform	low	5	68.1645	23.782479	95.951	reference
37	uniform	middle	15	70.3845	49.127608	305.268	reference
37	uniform	high	30	70.6556	98.116899	447.712	reference
37	labor	low	8	69.4196	28.120279	126.723	True
37	labor	middle	12	70.2734	35.438394	195.771	True
37	labor	high	36	70.6757	90.896695	435.429	True

PubMed LABOR K6 is the same operating point at all three tier comparisons. The arxiv/SAGE low-tier LABOR point is reported despite its infeasible calibration match. Actual final times need

not match the calibration target.

S5.1.2 Middle-budget four corners: accuracy, native margin and transitions

Table S5: HGCN / ogbn-arxiv: Middle-budget four corners: accuracy, native margin and transitions. Original displayed values, with all seeds retained.

Seed	Method	Corner	Accuracy %	Native margin	Fixed/repeat	Introduced/ repeat
11	uniform	sampled	70.1603	1.031111	baseline	baseline
11	uniform	local_restored	70.3408	1.037387	283.25	195.50
11	uniform	upstream_restored	70.3614	1.072653	1090.50	992.75
11	uniform	both_restored	70.4761	1.078604	1198.75	1045.25
11	labor	sampled	69.8337	0.989599	baseline	baseline
11	labor	local_restored	70.1232	1.015133	464.00	323.25
11	labor	upstream_restored	70.2626	1.055035	1451.25	1242.75
11	labor	both_restored	70.4427	1.075509	1601.75	1305.75
23	uniform	sampled	69.9854	1.185832	baseline	baseline
23	uniform	local_restored	70.2066	1.192160	324.25	216.75
23	uniform	upstream_restored	70.0384	1.241849	1165.25	1139.50
23	uniform	both_restored	70.2292	1.247544	1295.75	1177.25
23	labor	sampled	69.4407	1.123544	baseline	baseline
23	labor	local_restored	69.6912	1.153314	493.50	371.75
23	labor	upstream_restored	70.0116	1.223060	1629.75	1352.25
23	labor	both_restored	70.2426	1.243799	1784.00	1394.25
37	uniform	sampled	70.4113	1.116119	baseline	baseline
37	uniform	local_restored	70.5563	1.122096	281.75	211.25
37	uniform	upstream_restored	70.5903	1.147765	885.00	798.00
37	uniform	both_restored	70.7019	1.153284	1009.00	867.75
37	labor	sampled	70.0559	1.052758	baseline	baseline
37	labor	local_restored	70.3383	1.085157	462.75	325.50
37	labor	upstream_restored	70.4776	1.126096	1241.50	1036.50
37	labor	both_restored	70.7148	1.152714	1427.75	1107.50

S5.1.3 Paired restoration changes and interaction

Table S6: HGCN / ogbn-arxiv: Paired restoration changes and interaction. Original displayed values, with all seeds retained.

Seed	Method	Local pp	Upstream pp	Joint pp	Interaction pp	Local estimator squared L2
11	uniform	+0.1805	+0.2011	+0.3158	-0.0658	0.46850254
11	labor	+0.2896	+0.4290	+0.6090	-0.1096	4.61257300
23	uniform	+0.2212	+0.0530	+0.2438	-0.0303	0.46521756
23	labor	+0.2505	+0.5710	+0.8019	-0.0195	5.28277477
37	uniform	+0.1451	+0.1790	+0.2906	-0.0334	0.43290270
37	labor	+0.2824	+0.4218	+0.6589	-0.0453	3.70722188

S5.1.4 Exact computation and paid RAM reuse

Table S7: HGCN / ogbn-arxiv: Exact computation and paid RAM reuse. Original displayed values, with all seeds retained.

Seed	Exact accuracy %	q	Exact recompute s	Paid RAM s	RAM cache bytes
11	70.5553	1	0.265433	0.266946	27094880
11	70.5553	10	0.888393	0.349911	27094880
11	70.5553	100	7.117990	1.179566	27094880
23	70.1356	1	0.221386	0.227289	27094880
23	70.1356	10	0.771857	0.296584	27094880
23	70.1356	100	6.276569	0.989537	27094880
37	70.7179	1	0.229909	0.237918	27094880
37	70.7179	10	0.806239	0.319460	27094880
37	70.7179	100	6.569535	1.134879	27094880

These are the original frozen amortization formulas, not q extra experiments. Common CP/graph setup is separate; exact q=1 includes format setup. All full-memory, component timing, sampled-input/full-input errors, native margin changes, across-seed standard deviations and plan-level scalar records are retained in the exact CSV copies.

S5.2 HyboNet / ogbn-arxiv

S5.2.1 Deployment: all tiers and seeds

Table S8: HyboNet / ogbn-arxiv: Deployment: all tiers and seeds. Original displayed values, with all seeds retained.

Seed	Method	Tier	K	Accuracy %	Workload s	Max observed peak MiB	Feasible calibration match
11	uniform	low	5	67.3369	17.914019	61.660	reference
11	uniform	middle	15	69.9293	42.531143	159.404	reference
11	uniform	high	30	70.1325	89.502669	231.361	reference
11	labor	low	5	65.6060	17.973554	53.379	True
11	labor	middle	15	69.8120	36.009196	132.019	True
11	labor	high	36	70.1227	82.586572	225.251	True
23	uniform	low	5	67.1929	17.887291	61.666	reference
23	uniform	middle	15	69.8398	42.328543	160.632	reference
23	uniform	high	30	70.0420	89.328723	231.295	reference
23	labor	low	5	65.2990	17.745708	53.352	True
23	labor	middle	15	69.7565	35.834023	132.552	True
23	labor	high	36	70.0116	82.559198	224.706	True
37	uniform	low	5	67.5401	17.884926	61.700	reference
37	uniform	middle	15	70.2045	42.226411	160.698	reference
37	uniform	high	30	70.2734	89.086853	231.430	reference
37	labor	low	5	65.5993	17.750192	53.591	True
37	labor	middle	15	70.0934	35.836794	131.106	True

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Seed	Method	Tier	K	Accuracy %	Workload s	Max observed peak MiB	Feasible calibration match
37	labor	high	36	70.2734	82.519964	226.162	True

PubMed LABOR K6 is the same operating point at all three tier comparisons. The arxiv/SAGE low-tier LABOR point is reported despite its infeasible calibration match. Actual final times need not match the calibration target.

S5.2.2 Middle-budget four corners: accuracy, native margin and transitions

Table S9: HyboNet / ogbn-arxiv: Middle-budget four corners: accuracy, native margin and transitions. Original displayed values, with all seeds retained.

Seed	Method	Corner	Accuracy %	Native margin	Fixed/repeat	Introduced/ repeat
11	uniform	sampled	70.0111	1.357933	baseline	baseline
11	uniform	local_restored	70.1546	1.360817	251.75	182.00
11	uniform	upstream_restored	70.0224	1.428544	581.00	575.50
11	uniform	both_restored	70.1294	1.431504	711.50	654.00
11	labor	sampled	69.8604	1.350667	baseline	baseline
11	labor	local_restored	69.9664	1.355820	242.00	190.50
11	labor	upstream_restored	70.0363	1.427775	687.75	602.25
11	labor	both_restored	70.1294	1.431504	803.25	672.50
23	uniform	sampled	69.8074	1.298260	baseline	baseline
23	uniform	local_restored	69.9412	1.301650	248.00	183.00
23	uniform	upstream_restored	69.9566	1.344578	563.75	491.25
23	uniform	both_restored	70.0553	1.347913	699.25	578.75
23	labor	sampled	69.7436	1.300074	baseline	baseline
23	labor	local_restored	69.8712	1.305426	253.25	191.25
23	labor	upstream_restored	69.9458	1.345025	685.00	586.75
23	labor	both_restored	70.0553	1.347913	808.75	657.25
37	uniform	sampled	70.1983	1.294590	baseline	baseline
37	uniform	local_restored	70.2812	1.297774	235.00	194.75
37	uniform	upstream_restored	70.2431	1.350775	544.75	523.00
37	uniform	both_restored	70.3105	1.353502	665.00	610.50
37	labor	sampled	70.0847	1.281594	baseline	baseline
37	labor	local_restored	70.1834	1.284571	248.00	200.00
37	labor	upstream_restored	70.2163	1.352480	698.00	634.00
37	labor	both_restored	70.3105	1.353502	811.00	701.25

S5.2.3 Paired restoration changes and interaction

Table S10: HyboNet / ogbn-arxiv: Paired restoration changes and interaction. Original displayed values, with all seeds retained.

Seed	Method	Local pp	Upstream pp	Joint pp	Interaction pp	Local estimator squared L2
11	uniform	+0.1435	+0.0113	+0.1183	-0.0365	0.00224604
11	labor	+0.1060	+0.1759	+0.2690	-0.0129	0.03977844
23	uniform	+0.1337	+0.1492	+0.2479	-0.0350	0.00219455
23	labor	+0.1276	+0.2021	+0.3117	-0.0180	0.04161711
37	uniform	+0.0828	+0.0448	+0.1121	-0.0154	0.00211417
37	labor	+0.0988	+0.1317	+0.2258	-0.0046	0.03888045

S5.2.4 Exact computation and paid RAM reuse

Table S11: HyboNet / ogbn-arxiv: Exact computation and paid RAM reuse. Original displayed values, with all seeds retained.

Seed	Exact accuracy %	q	Exact recompute s	Paid RAM s	RAM cache bytes
11	70.1294	1	0.340264	0.334069	27094880
11	70.1294	10	1.830769	0.409512	27094880
11	70.1294	100	16.735813	1.163942	27094880
23	70.0553	1	0.284210	0.289640	27094880
23	70.0553	10	1.550985	0.356633	27094880
23	70.0553	100	14.218742	1.026562	27094880
37	70.3105	1	0.393480	0.382282	27094880
37	70.3105	10	1.984050	0.446529	27094880
37	70.3105	100	17.889743	1.088996	27094880

These are the original frozen amortization formulas, not q extra experiments. Common CP/graph setup is separate; exact q=1 includes format setup. All full-memory, component timing, sampled-input/full-input errors, native margin changes, across-seed standard deviations and plan-level scalar records are retained in the exact CSV copies.

S5.3 GraphSAGE / ogbn-arxiv

S5.3.1 Deployment: all tiers and seeds

Table S12: GraphSAGE / ogbn-arxiv: Deployment: all tiers and seeds. Original displayed values, with all seeds retained.

Seed	Method	Tier	K	Accuracy %	Workload s	Max observed peak MiB	Feasible calibration match
11	uniform	low	5	69.0559	18.178637	51.506	reference
11	uniform	middle	15	70.4838	43.045747	108.159	reference

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Seed	Method	Tier	K	Accuracy %	Workload s	Max observed peak MiB	Feasible calibration match
11	uniform	high	30	70.6078	89.902420	165.709	reference
11	labor	low	5	67.9865	18.437689	47.683	False
11	labor	middle	15	70.4884	36.302164	91.150	True
11	labor	high	36	70.5805	82.874647	154.440	True
23	uniform	low	5	69.3666	18.222668	51.504	reference
23	uniform	middle	15	70.7497	43.126781	107.987	reference
23	uniform	high	30	70.8624	90.123444	165.785	reference
23	labor	low	5	68.2869	18.421956	47.648	False
23	labor	middle	15	70.7369	36.452571	91.200	True
23	labor	high	36	70.8809	83.086254	154.961	True
37	uniform	low	5	69.8969	18.512795	51.506	reference
37	uniform	middle	15	71.3408	43.432314	108.366	reference
37	uniform	high	30	71.4045	91.046472	165.621	reference
37	labor	low	5	69.0148	18.442969	47.800	False
37	labor	middle	15	71.2528	36.686506	90.864	True
37	labor	high	36	71.3382	84.378858	154.859	True

PubMed LABOR K6 is the same operating point at all three tier comparisons. The arxiv/SAGE low-tier LABOR point is reported despite its infeasible calibration match. Actual final times need not match the calibration target.

S5.3.2 Middle-budget four corners: accuracy, native margin and transitions

Table S13: GraphSAGE / ogbn-arxiv: Middle-budget four corners: accuracy, native margin and transitions. Original displayed values, with all seeds retained.

Seed	Method	Corner	Accuracy %	Native margin	Fixed/repeat	Introduced/ repeat
11	uniform	sampled	70.5327	1.252588	baseline	baseline
11	uniform	local_restored	70.5970	1.256131	186.50	155.25
11	uniform	upstream_restored	70.5641	1.312539	605.00	589.75
11	uniform	both_restored	70.6232	1.315913	674.50	630.50
11	labor	sampled	70.4169	1.254852	baseline	baseline
11	labor	local_restored	70.5286	1.257397	198.75	144.50
11	labor	upstream_restored	70.5646	1.313823	697.75	626.00
11	labor	both_restored	70.6232	1.315913	773.50	673.25
23	uniform	sampled	70.7353	1.258992	baseline	baseline
23	uniform	local_restored	70.8331	1.262131	201.75	154.25
23	uniform	upstream_restored	70.7899	1.319129	648.75	622.25
23	uniform	both_restored	70.8804	1.322171	728.50	658.00
23	labor	sampled	70.6916	1.260396	baseline	baseline
23	labor	local_restored	70.7688	1.262875	202.25	164.75
23	labor	upstream_restored	70.7914	1.321024	712.25	663.75
23	labor	both_restored	70.8804	1.322171	790.75	699.00
37	uniform	sampled	71.2898	1.216943	baseline	baseline
37	uniform	local_restored	71.3505	1.219746	195.50	166.00

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Seed	Method	Corner	Accuracy %	Native margin	Fixed/repeat	Introduced/ repeat
37	uniform	upstream_restored	71.3634	1.269457	559.75	524.00
37	uniform	both_restored	71.4051	1.271965	640.00	584.00
37	labor	sampled	71.3022	1.208778	baseline	baseline
37	labor	local_restored	71.4128	1.210714	204.50	150.75
37	labor	upstream_restored	71.3274	1.269505	599.25	587.00
37	labor	both_restored	71.4051	1.271965	686.50	636.50

S5.3.3 Paired restoration changes and interaction

Table S14: GraphSAGE / ogbn-arxiv: Paired restoration changes and interaction. Original displayed values, with all seeds retained.

Seed	Method	Local pp	Upstream pp	Joint pp	Interaction pp	Local estimator squared L2
11	uniform	+0.0643	+0.0314	+0.0905	-0.0051	0.00352917
11	labor	+0.1116	+0.1476	+0.2063	-0.0530	0.00385210
23	uniform	+0.0977	+0.0545	+0.1451	-0.0072	0.00361693
23	labor	+0.0772	+0.0998	+0.1888	+0.0118	0.00399701
37	uniform	+0.0607	+0.0736	+0.1152	-0.0190	0.00724469
37	labor	+0.1106	+0.0252	+0.1029	-0.0329	0.00779397

S5.3.4 Exact computation and paid RAM reuse

Table S15: GraphSAGE / ogbn-arxiv: Exact computation and paid RAM reuse. Original displayed values, with all seeds retained.

Seed	Exact accuracy %	q	Exact recompute s	Paid RAM s	RAM cache bytes
11	70.6232	1	0.187096	0.193923	27094880
11	70.6232	10	0.444067	0.267763	27094880
11	70.6232	100	3.013769	1.006162	27094880
23	70.8804	1	0.177334	0.184627	27094880
23	70.8804	10	0.436640	0.262394	27094880
23	70.8804	100	3.029695	1.040062	27094880
37	71.4051	1	0.187917	0.195670	27094880
37	71.4051	10	0.455764	0.277773	27094880
37	71.4051	100	3.134234	1.098801	27094880

These are the original frozen amortization formulas, not q extra experiments. Common CP/graph setup is separate; exact q=1 includes format setup. All full-memory, component timing, sampled-input/full-input errors, native margin changes, across-seed standard deviations and plan-level scalar records are retained in the exact CSV copies.

S5.4 GAT / ogbn-arxiv

S5.4.1 Deployment: all tiers and seeds

Table S16: GAT / ogbn-arxiv: Deployment: all tiers and seeds. Original displayed values, with all seeds retained.

Seed	Method	Tier	K	Accuracy %	Workload s	Max observed peak MiB	Feasible calibration match
11	uniform	low	5	70.8089	18.178965	197.702	reference
11	uniform	middle	15	72.0578	44.617388	1251.475	reference
11	uniform	high	30	72.2322	94.091825	3080.454	reference
11	labor	low	5	70.3902	18.303514	169.991	True
11	labor	middle	15	72.0053	37.731604	943.442	True
11	labor	high	30	72.1797	74.619808	2398.835	True
23	uniform	low	5	71.0501	18.486565	197.854	reference
23	uniform	middle	15	72.2219	45.028231	1251.033	reference
23	uniform	high	30	72.3417	95.111255	3081.372	reference
23	labor	low	5	70.6109	18.731157	167.620	True
23	labor	middle	15	72.1473	38.313267	946.144	True
23	labor	high	30	72.3808	75.057174	2404.393	True
37	uniform	low	5	71.2667	18.293175	198.204	reference
37	uniform	middle	15	72.1987	44.807298	1260.540	reference
37	uniform	high	30	72.1890	94.699773	3076.838	reference
37	labor	low	5	70.8881	18.345737	170.414	True
37	labor	middle	15	72.1437	37.926430	929.912	True
37	labor	high	30	72.2373	75.009215	2400.219	True

PubMed LABOR K6 is the same operating point at all three tier comparisons. The arxiv/SAGE low-tier LABOR point is reported despite its infeasible calibration match. Actual final times need not match the calibration target.

S5.4.2 Middle-budget four corners: accuracy, native margin and transitions

Table S17: GAT / ogbn-arxiv: Middle-budget four corners: accuracy, native margin and transitions. Original displayed values, with all seeds retained.

Seed	Method	Corner	Accuracy %	Native margin	Fixed/repeat	Introduced/ repeat
11	uniform	sampled	72.0501	3.445456	baseline	baseline
11	uniform	local_restored	72.1442	3.451363	181.75	136.00
11	uniform	upstream_restored	72.2049	3.680112	441.50	366.25
11	uniform	both_restored	72.2774	3.685698	532.25	421.75
11	labor	sampled	72.0130	3.441008	baseline	baseline
11	labor	local_restored	72.1036	3.448998	188.25	144.25
11	labor	upstream_restored	72.2126	3.680089	523.25	426.25
11	labor	both_restored	72.2774	3.685698	609.00	480.50

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Seed	Method	Corner	Accuracy %	Native margin	Fixed/repeat	Introduced/ repeat
23	uniform	sampled	72.2059	3.461650	baseline	baseline
23	uniform	local_restored	72.2851	3.466644	180.50	142.00
23	uniform	upstream_restored	72.3176	3.719926	429.00	374.75
23	uniform	both_restored	72.3885	3.725181	527.75	439.00
23	labor	sampled	72.2167	3.464314	baseline	baseline
23	labor	local_restored	72.3160	3.465955	183.75	135.50
23	labor	upstream_restored	72.3021	3.726340	490.50	449.00
23	labor	both_restored	72.3885	3.725181	574.00	490.50
37	uniform	sampled	72.1684	3.517426	baseline	baseline
37	uniform	local_restored	72.2332	3.522991	174.00	142.50
37	uniform	upstream_restored	72.1756	3.732949	422.00	418.50
37	uniform	both_restored	72.2198	3.737844	509.00	484.00
37	labor	sampled	72.0979	3.486986	baseline	baseline
37	labor	local_restored	72.1432	3.487776	181.50	159.50
37	labor	upstream_restored	72.1746	3.734061	495.25	458.00
37	labor	both_restored	72.2198	3.737844	578.25	519.00

S5.4.3 Paired restoration changes and interaction

Table S18: GAT / ogbn-arxiv: Paired restoration changes and interaction. Original displayed values, with all seeds retained.

Seed	Method	Local pp	Upstream pp	Joint pp	Interaction pp	Local estimator squared L2
11	uniform	+0.0941	+0.1548	+0.2274	-0.0216	1.90133056
11	labor	+0.0905	+0.1996	+0.2644	-0.0257	2.04853182
23	uniform	+0.0792	+0.1116	+0.1826	-0.0082	2.34209336
23	labor	+0.0993	+0.0854	+0.1718	-0.0129	2.66721869
37	uniform	+0.0648	+0.0072	+0.0514	-0.0206	2.05886113
37	labor	+0.0453	+0.0766	+0.1219	+0.0000	2.31235536

S5.4.4 Exact computation and paid RAM reuse

Table S19: GAT / ogbn-arxiv: Exact computation and paid RAM reuse. Original displayed values, with all seeds retained.

Seed	Exact accuracy %	q	Exact recompute s	Paid RAM s	RAM cache bytes
11	72.2774	1	0.307020	0.434544	27094880
11	72.2774	10	1.998912	1.704486	27094880
11	72.2774	100	18.917833	14.403904	27094880
23	72.3885	1	0.297910	0.304950	27094880
23	72.3885	10	1.859950	0.377374	27094880
23	72.3885	100	17.480343	1.101615	27094880
37	72.2198	1	0.292912	0.299229	27094880

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Seed	Exact accuracy %	q	Exact recompute s	Paid RAM s	RAM cache bytes
37	72.2198	10	1.851666	0.365270	27094880
37	72.2198	100	17.439200	1.025682	27094880

These are the original frozen amortization formulas, not q extra experiments. Common CP/graph setup is separate; exact q=1 includes format setup. All full-memory, component timing, sampled-input/full-input errors, native margin changes, across-seed standard deviations and plan-level scalar records are retained in the exact CSV copies.

S5.5 GraphSAGE / PubMed

S5.5.1 Deployment: all tiers and seeds

Table S20: GraphSAGE / PubMed: Deployment: all tiers and seeds. Original displayed values, with all seeds retained.

Seed	Method	Tier	K	Accuracy %	Workload s	Max observed peak MiB	Feasible calibration match
11	uniform	low	1	72.8500	0.297638	37.762	reference
11	uniform	middle	4	74.7250	0.329297	48.422	reference
11	uniform	high	8	75.0000	0.387821	63.431	reference
11	labor	low	6	75.1250	0.360889	54.098	True
11	labor	middle	6	75.1250	0.360421	54.098	True
11	labor	high	6	75.1250	0.364992	54.098	True
23	uniform	low	1	73.6250	0.285209	37.772	reference
23	uniform	middle	4	75.5500	0.326600	48.411	reference
23	uniform	high	8	75.8250	0.377677	62.183	reference
23	labor	low	6	75.5750	0.351006	54.274	True
23	labor	middle	6	75.5750	0.345077	54.274	True
23	labor	high	6	75.5750	0.330342	54.274	True
37	uniform	low	1	73.5750	0.285183	37.771	reference
37	uniform	middle	4	74.3000	0.330403	48.230	reference
37	uniform	high	8	74.3250	0.361659	63.245	reference
37	labor	low	6	74.2250	0.333999	53.901	True
37	labor	middle	6	74.2250	0.332198	53.901	True
37	labor	high	6	74.2250	0.339028	53.901	True

PubMed LABOR K6 is the same operating point at all three tier comparisons. The arxiv/SAGE low-tier LABOR point is reported despite its infeasible calibration match. Actual final times need not match the calibration target.

S5.5.2 Middle-budget four corners: accuracy, native margin and transitions

Table S21: GraphSAGE / PubMed: Middle-budget four corners: accuracy, native margin and transitions. Original displayed values, with all seeds retained.

Seed	Method	Corner	Accuracy %	Native margin	Fixed/repeat	Introduced/ repeat
11	uniform	sampled	74.8000	5.378680	baseline	baseline
11	uniform	local_restored	74.7500	5.378052	2.75	3.25
11	uniform	upstream_restored	75.3750	5.422523	18.75	13.00
11	uniform	both_restored	75.8000	5.421434	21.75	11.75
11	labor	sampled	75.0500	5.384160	baseline	baseline
11	labor	local_restored	75.1750	5.380556	2.00	0.75
11	labor	upstream_restored	75.2500	5.424648	14.75	12.75
11	labor	both_restored	75.8000	5.421434	18.00	10.50
23	uniform	sampled	75.2250	5.597354	baseline	baseline
23	uniform	local_restored	75.2750	5.589564	3.25	2.75
23	uniform	upstream_restored	75.2250	5.672130	13.75	13.75
23	uniform	both_restored	75.4000	5.661963	16.50	14.75
23	labor	sampled	75.4250	5.589689	baseline	baseline
23	labor	local_restored	75.5750	5.598738	3.25	1.75
23	labor	upstream_restored	75.2250	5.655098	9.25	11.25
23	labor	both_restored	75.4000	5.661963	10.75	11.00
37	uniform	sampled	74.3000	5.621925	baseline	baseline
37	uniform	local_restored	74.1750	5.616949	2.75	4.00
37	uniform	upstream_restored	74.5500	5.638979	15.00	12.50
37	uniform	both_restored	74.6000	5.635255	17.25	14.25
37	labor	sampled	73.9750	5.585634	baseline	baseline
37	labor	local_restored	73.9750	5.594271	1.75	1.75
37	labor	upstream_restored	74.5750	5.628201	15.75	9.75
37	labor	both_restored	74.6000	5.635255	15.75	9.50

S5.5.3 Paired restoration changes and interaction

Table S22: GraphSAGE / PubMed: Paired restoration changes and interaction. Original displayed values, with all seeds retained.

Seed	Method	Local pp	Upstream pp	Joint pp	Interaction pp	Local estimator squared L2
11	uniform	-0.0500	+0.5750	+1.0000	+0.4750	0.43419411
11	labor	+0.1250	+0.2000	+0.7500	+0.4250	0.25247188
23	uniform	+0.0500	+0.0000	+0.1750	+0.1250	0.40992959
23	labor	+0.1500	-0.2000	-0.0250	+0.0250	0.23164653
37	uniform	-0.1250	+0.2500	+0.3000	+0.1750	0.38037591
37	labor	+0.0000	+0.6000	+0.6250	+0.0250	0.22698362

S5.5.4 Exact computation and paid RAM reuse

Table S23: GraphSAGE / PubMed: Exact computation and paid RAM reuse. Original displayed values, with all seeds retained.

Seed	Exact accuracy %	q	Exact recompute s	Paid RAM s	RAM cache bytes
11	75.8000	1	0.085218	0.085334	236604
11	75.8000	10	0.155151	0.087565	236604
11	75.8000	100	0.854481	0.109872	236604
23	75.4000	1	0.078141	0.078283	236604
23	75.4000	10	0.135458	0.080583	236604
23	75.4000	100	0.708623	0.103591	236604
37	74.6000	1	0.075179	0.075279	236604
37	74.6000	10	0.131587	0.077243	236604
37	74.6000	100	0.695659	0.096879	236604

These are the original frozen amortization formulas, not q extra experiments. Common CP/graph setup is separate; exact q=1 includes format setup. All full-memory, component timing, sampled-input/full-input errors, native margin changes, across-seed standard deviations and plan-level scalar records are retained in the exact CSV copies.

S6 Exact Computation and RAM Reuse

The paid reuse figure is moved intact from the main paper. It evaluates the registered formulas at $q = 1, 10, 100$ from the recorded timing medians, not from additional repeated-workload experiments. Format setup and full-output cache construction, including device-to-host transfer, are paid before RAM gathering. Common graph/checkpoint loading remains separately scoped. A fixed graph and model are required; changes invalidate the cache. In particular, a cached full-graph output is not a free hidden representation or a teacher input to the sampled predictor.

The arxiv/SAGE mean paid RAM time is 0.191 seconds at $q = 1$, compared with 0.184 seconds for exact recomputation, and 0.269 versus 0.445 seconds at $q = 10$. GAT's $q = 100$ paid times of 14.404, 1.102 and 1.026 seconds retain their seed-level variation and host-context limitation. These endpoints do not establish an interpolated break-even guarantee. Main-paper memory comparisons are scoped to the displayed middle tier: a high-budget sampled GAT point can use more allocated GPU memory than exact computation.

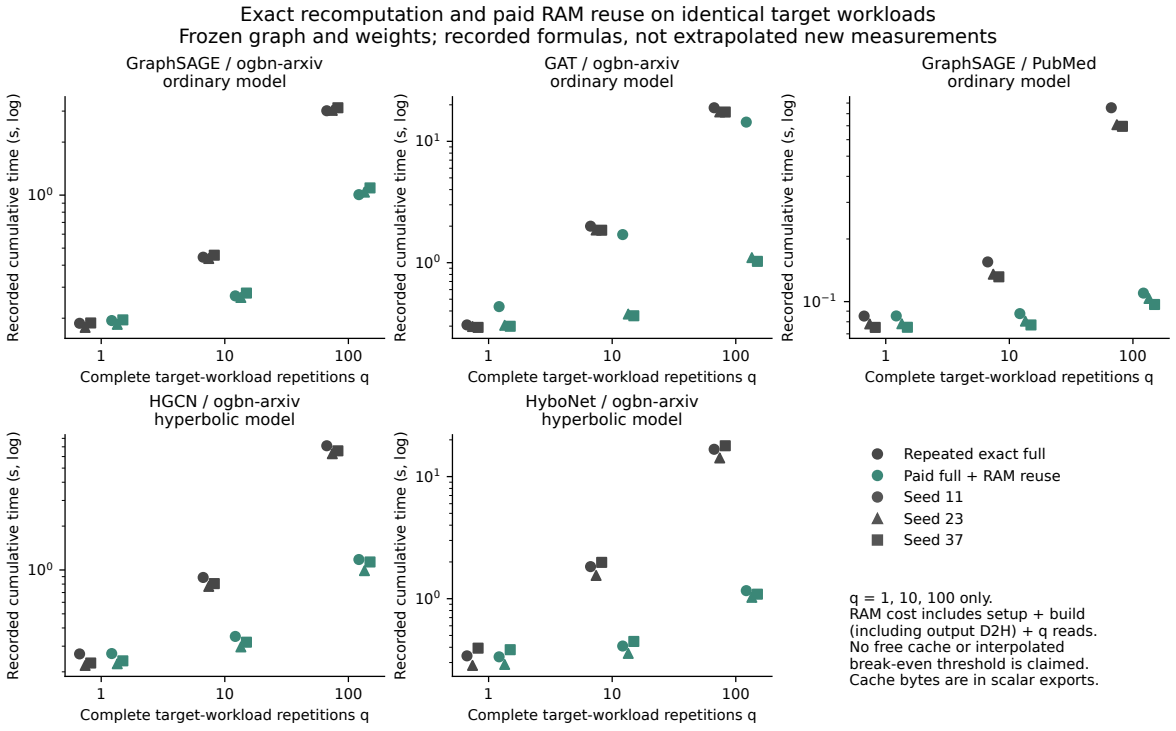


Figure S1: Registered exact/RAM amortization at $q = 1, 10, 100$ repetitions of the identical complete target workload. Exact recomputation pays format setup plus q full-forward/D2H/target-gather costs; RAM reuse pays format setup plus one full cache build including D2H, followed by q RAM gathers. Common graph/checkpoint setup is separately reported. These are the recorded frozen formulas using the original timing medians, not q new experiments or an interpolated break-even estimate. The graph and weights must remain fixed; CPU cache size is retained in the accompanying scalar export.

S7 Historical Evidence and Auxiliary Cases

The accepted offline Task A synthesis provides a training-regime contrast that the new frozen-model study deliberately does not repeat. It covers 24 checkpoints and 16 original fit plans per checkpoint, retaining all three seeds. For HyboNet/Airport, the full-trained four-corner accuracies (sampled, local, upstream, both) are 79.1141%, 79.8407%, 92.6383% and 93.4174%; the matched-trained values are 88.8480%, 89.3557%, 89.4914% and 89.9860%. Upstream restoration therefore changes from +13.5242 to +0.6434 points, while local restoration changes from +0.7265 to +0.5077. The checkpoints differ between training regimes. This is descriptive development evidence about different learned functions, not a same-weight causal training intervention or an independent final-test result.

On Disease-NC, HyboNet's full-trained four-corner means are 80.9524%, 83.3829%, 91.4187% and 95.2381%; matched means are 76.1905%, 77.2321%, 77.8770% and 79.3651%. The historical HGCN has only one message layer, so upstream restoration is a structural alias of sampled computation and joint restoration aliases local restoration. Its Disease-NC matched mean falls from 63.2937% sampled to 62.6984% under local restoration. Seed 11 has 11 corrections and 23 harms over 672 node–plan events, despite a positive mean-margin change; the degenerate seed 37 remains at 50%. These cases are retained and do not describe the new three-layer HGCN.

Predictive explanation and correction remain bounded. Earlier held-out explanatory prediction tests obtained geometric/operator-feature cross-entropy reductions of 2.8669% and 1.1289%, below the registered 5% criterion; margin-MSE and calibration criteria also failed. This study does not refit them. The HyboNet E1 correction scored 89.5658%, compared with 89.5965% for ordinary averaging of the same two native-logit views and 89.0275% for one view. Among 143 differing predictions, E1 corrected 61 and harmed 68; a single-view gain did not establish incremental value over the equal-information control.

A separate HGCN/arxiv correction confirmation gives K15 means of 71.2164%, 71.2880% and 71.2220% for plain, candidate and generic control, respectively. Candidate changes across seeds are −0.0906, +0.1292 and +0.1762 points, while request time rises from 118.174 to 155.623 ms and allocated memory from 303 to 1,019 MiB. These adapted weights are not substituted into the new original-plain comparison. Conditional positive WordNet results belong to a different operator/task/development contract and do not reverse these limitations. Complete historical tables, plots, controls, calibration failures and source references are retained in the indexed companion historical files, including the original SAGE gate and exact/RAM records; no historical sign or seed is removed to support the new narrative.

S8 Reproducibility and Complete Record Index

The final scalar synthesis is accepted under SHA256:
`09bd47fbc1ad872f61c155b901a39da9d3c4600e69339d5b2b693ce8d9a36f05`

It retains 2,115 per-seed scalar rows and 705 across-seed rows for deployment/cost, plus 1,440 per-seed and 480 across-seed mechanism rows, all prescribed scalar repeats and 120 error–decision pairs. The tables above format these accepted summaries; decimal units are displayed as in the prior readable tables. They are not new statistical estimates.

- `reproducibility/final-accepted-scalars/`: original-precision accepted scalar exports and index.

- `reproducibility/final-per-seed-results.md`: readable final seed-level source transcribed into this supplement.
- `reproducibility/a800-budget-locks-v1.tex` and companion CSV/JSON: complete original label-free calibration, including infeasible and nonmonotone observations.
- `reproducibility/task-a-findings.json`, `task-a-cell-summary.csv` and their canonical source links: historical 24-checkpoint fit synthesis, preserving original event denominators. The original per-checkpoint condition/contrast/interaction exports supply every historical seed, rather than a pooled mean replacing it.
- `reproducibility/historical-evidence-full.tex`: preserved earlier historical narrative, including narrower prior subsets, WordNet controls and calibration/predictor failures. Its dated scope and pending language are not a current final-status assertion.
- `reproducibility/math-implementation-map.md`: exact file/line/source hashes, including the native Lorentz denominator, prototype, bias and distance-proxy conditions. Its current final-status statements distinguish accepted empirical evidence from unchanged conditional mathematical contracts.
- `reproducibility/literature-and-formula-attribution.md`: verified primary references, elementary identities and this study's diagnostic/empirical contribution boundaries. The companion `reproducibility/supplement-package-plan.md` distinguishes reading material, scalar/plot reproduction and full experimental reproduction.
- `reproducibility/portable-proposal/`: receipt-based runtime versions, native revisions, official data acquisition/preprocessing, license gaps and the proposed portable export boundary.

The new round's allocated GPU use is 17,299 seconds: 269 for three PubMed base trainings, 311 for successful label-free calibration, 13 for the preserved metadata failure and 16,706 for final evaluation. All resources are closed. The metadata repair changed no model, sampler or numerical computation, and accepted qualifications were not repeated. Allocation time is not deployment latency. Nested scheduler, worker, controller and phase timings have different or overlapping scopes and are not added; missing child CPU observations are not replaced with zero. The final offline synthesis records 11.375 native CPU-seconds and 14.666 wall-seconds without added GPU computation.

No checkpoint bytes, task arrays, labels, credentials or private scheduler connection details are bundled in this manuscript directory. Exact native-source reconstruction, redistribution rights, reviewer-accessible checkpoint availability and a portable entry point remain explicit completion items. These packaging questions require no outcome-driven experiment or rescoring.

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